Worksheet 3

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Section/Group: 3-A
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Subject Name: Internet of Things Lab
Subject Code: 24CAP-751

1. Aim/Overview of the practical: Design an access control system for exam hall.

Requirements:

1. Only registered students can enter.
2. Print:
    - Name
    - Roll Number
    - "Exam Access Granted"
3. For others:
    - Print "Not Eligible for Exam"

2. Apparatus

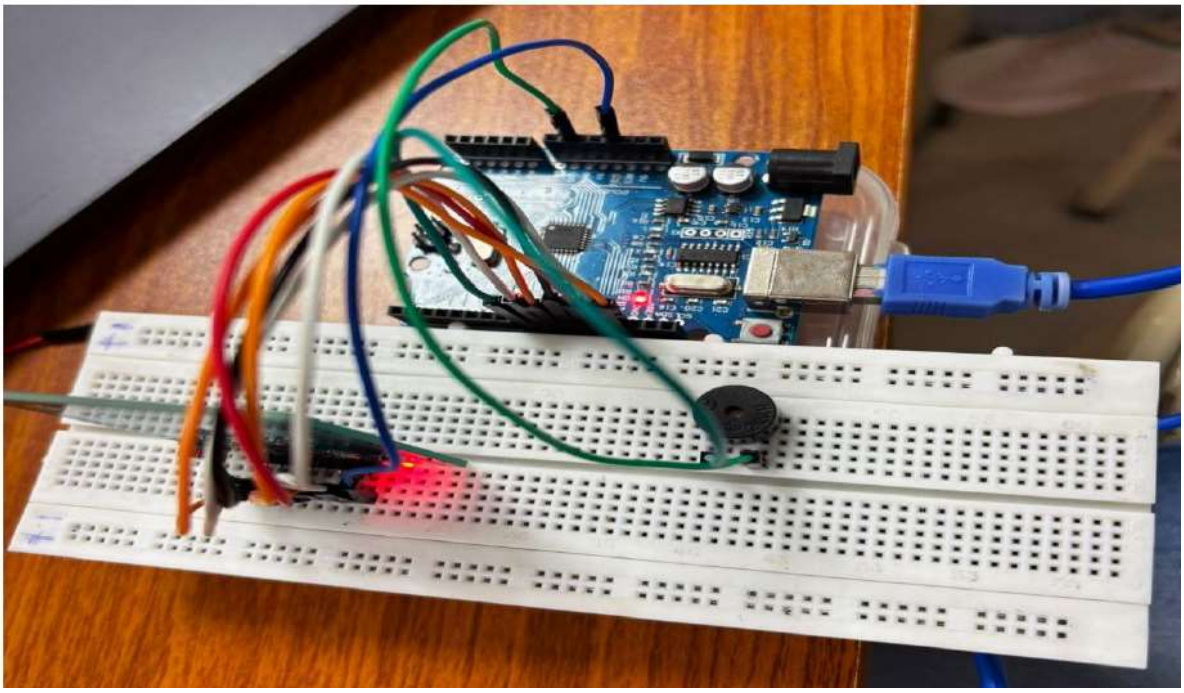
Hardware Requirements

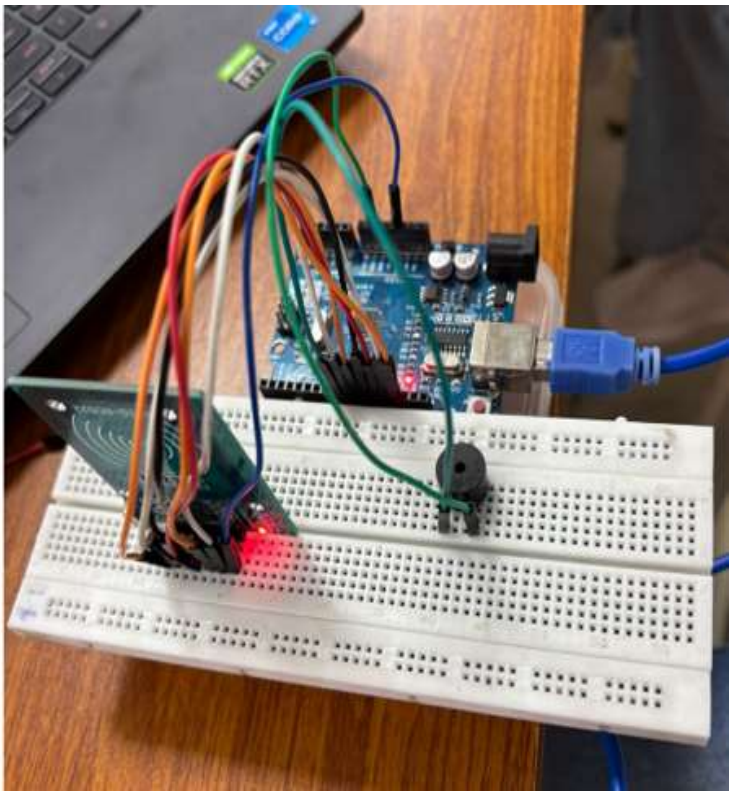
Breadboard
RFID Reader Module (RC522)
Connecting wires
Buzzer
USB Cable

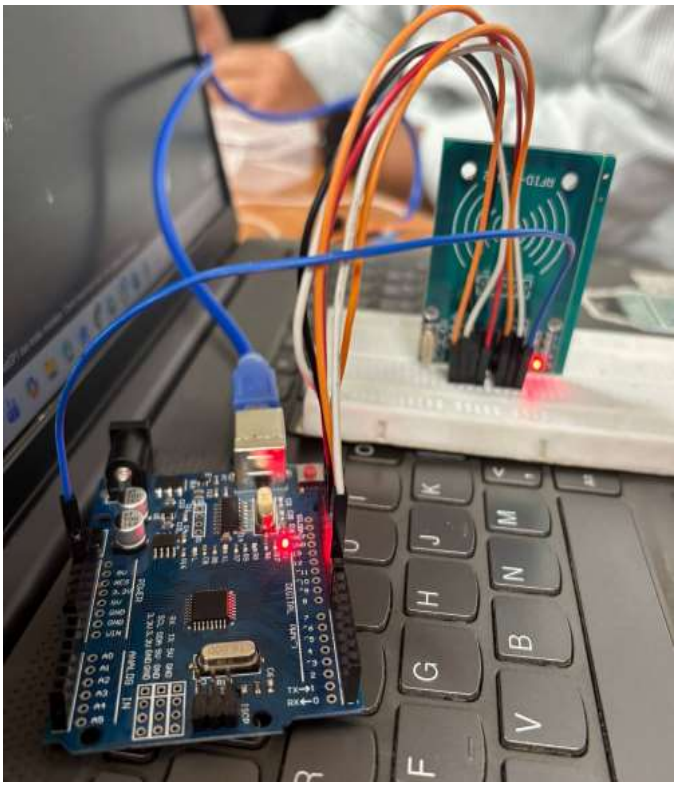
Software requirements

Arduino programming environment

3. Circuit Diagram(TinkerCad):







1. **Coding:**

Dumpinfo code: It is used to read and display the UID and full data of an RFID card on the serial monitor**.**

```
/*
 * ----------------------------------------------------------------------------------------------------
 * Example sketch/program showing how to read data from a PICC to serial.
 * ----------------------------------------------------------------------------------------------------
 * This is a MFRC522 library example; for further details and other examples see: https://github.com/miguelbalboa/rfid
 *
 * Example sketch/program showing how to read data from a PICC (that is: a RFID Tag or Card) using a MFRC522 based RFID
 * Reader on the Arduino SPI interface.
 *
 * When the Arduino and the MFRC522 module are connected (see the pin layout below), load this sketch into Arduino IDE
 * then verify/compile and upload it. To see the output: use Tools, Serial Monitor of the IDE (hit Ctrl+Shft+M). When
 * you present a PICC (that is: a RFID Tag or Card) at reading distance of the MFRC522 Reader/PCD, the serial output
 * will show the ID/UID, type and any data blocks it can read. Note: you may see "Timeout in communication" messages
 * when removing the PICC from reading distance too early.
 *
 * If your reader supports it, this sketch/program will read all the PICCs presented (that is: multiple tag reading).
 * So if you stack two or more PICCs on top of each other and present them to the reader, it will first output all
 * details of the first and then the next PICC. Note that this may take some time as all data blocks are dumped, so
```

```
 * keep the PICCs at reading distance until complete.
 *
 * @license Released into the public domain.
 *
 * Typical pin layout used:
 * -----------------------------------------------------------------------------------------
 *             MFRC522      Arduino       Arduino   Arduino    Arduino          Arduino
 *             Reader/PCD   Uno/101       Mega      Nano v3    Leonardo/Micro   Pro Micro
 * Signal      Pin          Pin           Pin       Pin        Pin              Pin
 * -----------------------------------------------------------------------------------------
 * RST/Reset   RST          9             5         D9         RESET/ICSP-5     RST
 * SPI SS      SDA(SS)      10            53        D10        10               10
 * SPI MOSI    MOSI         11 / ICSP-4   51        D11        ICSP-4           16
 * SPI MISO    MISO         12 / ICSP-1   50        D12        ICSP-1           14
 * SPI SCK     SCK          13 / ICSP-3   52        D13        ICSP-3           15
 *
 * More pin layouts for other boards can be found here:
https://github.com/miguelbalboa/rfid#pin-layout
 */


#include <SPI.h>
#include <MFRC522.h>

#define RST_PIN         9          // Configurable, see typical pin layout above
#define SS_PIN          10         // Configurable, see typical pin layout above

MFRC522 mfrc522(SS_PIN, RST_PIN);  // Create MFRC522 instance

void setup() {
  Serial.begin(9600);   // Initialize serial communications with the PC
  while (!Serial);    // Do nothing if no serial port is opened (added for Arduinos based on
ATMEGA32U4)
  SPI.begin();      // Init SPI bus
```

```
  mfrc522.PCD_Init();     // Init MFRC522
  delay(4);       // Optional delay. Some board do need more time after init to be
ready, see Readme

  mfrc522.PCD_DumpVersionToSerial();  // Show details of PCD - MFRC522 Card
Reader details

  Serial.println(F("Scan PICC to see UID, SAK, type, and data blocks..."));
}

void loop() {
  // Reset the loop if no new card present on the sensor/reader. This saves the
entire process when idle.
  if ( ! mfrc522.PICC_IsNewCardPresent()) {
    return;
  }

  // Select one of the cards
  if ( ! mfrc522.PICC_ReadCardSerial()) {
    return;
  }

  // Dump debug info about the card; PICC_HaltA() is automatically called
  mfrc522.PICC_DumpToSerial(&(mfrc522.uid));
}
```

Card UID

Output Serial Monitor ×

mm

```
Card UID: 4D 64 9D 67
Card SAK: 08
PICC type: MIFARE 1KB
Sector Block   0  1  2  3   4  5  6  7   8  9 10 11  12 13 14 15  AccessBits
  15     63   00 00 00 00  00 00 FF 07  80 69 FF FF  FF FF FF FF  [ 0 0 1 ]
         62   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
         61   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
         60   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
  14     59   00 00 00 00  00 00 FF 07  80 69 FF FF  FF FF FF FF  [ 0 0 1 ]
         58   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
         57   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
         56   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
  13     55   00 00 00 00  00 00 FF 07  80 69 FF FF  FF FF FF FF  [ 0 0 1 ]
         54   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
         53   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
         52   00 00 00 00  00 00 00 00  00 00 00 00  00 00 00 00  [ 0 0 0 ]
  12     51  PCD_Authenticate() failed: Timeout in communication.
  11     47  PCD_Authenticate() failed: Timeout in communication.
  10     43  PCD_Authenticate() failed: Timeout in communication.
   9     39  PCD_Authenticate() failed: Timeout in communication.
   8     35  PCD_Authenticate() failed: Timeout in communication.
   7     31  PCD_Authenticate() failed: Timeout in communication.
   6     27  PCD_Authenticate() failed: Timeout in communication.
   5     23  PCD_Authenticate() failed: Timeout in communication.
   4     19  PCD_Authenticate() failed: Timeout in communication.
```

This code checks, It is whether a scanned RFID card belongs to a registered student or not.

```
#include <SPI.h>            // SPI communication library
#include <MFRC522.h>        // RFID library

#define RST_PIN 9           //Reset pin
#define SS_PIN  10          // Slave Select pin
#define buzzer 8            // Buzzer pin

MFRC522 mfrc522(SS_PIN, RST_PIN);          // Create RFID object

//3 Registered Student UIDs
byte student1[4] = {0x4D, 0x64, 0x9D, 0x67};
byte student2[4] = {0xC9, 0x30, 0x71, 0x60};
byte student3[4] = {0xA9, 0xF7, 0x9D, 0x94};

void setup()
{
  Serial.begin(9600);            // Initialize serial communications with the PC
  pinMode(buzzer,OUTPUT);

  while (!Serial);
  SPI.begin();                   // Init SPI bus
  mfrc522.PCD_Init();            // Init MFRC522
  Serial.println("Exam Hall Access System Ready...");
}

void loop()
{
     // Check if new card is present and read UID
  if (mfrc522.PICC_IsNewCardPresent() && mfrc522.PICC_ReadCardSerial())
  {
```

```
Serial.println("Card Scanned...");

//Student 1
if(mfrc522.uid.uidByte[0] == student1[0] &&
   mfrc522.uid.uidByte[1] == student1[1] &&
   mfrc522.uid.uidByte[2] == student1[2] &&
   mfrc522.uid.uidByte[3] == student1[3])
{
  Serial.println("Name: Aaryaman Singh");
  Serial.println("Roll Number: 24MCA20449");
  Serial.println("Exam Access Granted");

  digitalWrite(buzzer,HIGH); delay(500);
  digitalWrite(buzzer,LOW);  delay(500);
}

//Student 2
else if(mfrc522.uid.uidByte[0] == student2[0] &&
        mfrc522.uid.uidByte[1] == student2[1] &&
        mfrc522.uid.uidByte[2] == student2[2] &&
        mfrc522.uid.uidByte[3] == student2[3])
{
  Serial.println("Name: Ankush Koundal");
  Serial.println("Roll Number: 24MCA20251");
  Serial.println("Exam Access Granted");

  digitalWrite(buzzer,HIGH); delay(500);
  digitalWrite(buzzer,LOW);  delay(500);
}

// Student 3
else if(mfrc522.uid.uidByte[0] == student3[0] &&
        mfrc522.uid.uidByte[1] == student3[1] &&
        mfrc522.uid.uidByte[2] == student3[2] &&
```

```
        mfrc522.uid.uidByte[3] == student3[3])
    {
      Serial.println("Name: Dimpal Tyagi");
      Serial.println("Roll Number: 24MCA20021");
      Serial.println("Exam Access Granted");

      digitalWrite(buzzer,HIGH); delay(500);
      digitalWrite(buzzer,LOW);  delay(500);
    }

    //Not Registered
    else
    {
      Serial.println("Not Eligible for Exam");

      digitalWrite(buzzer,HIGH); delay(200);
      digitalWrite(buzzer,LOW);  delay(200);
      digitalWrite(buzzer,HIGH); delay(200);
      digitalWrite(buzzer,LOW);  delay(200);
    }

    Serial.println("----------------------");

    mfrc522.PICC_HaltA();     //stop reading card
  }
}
```

UID matched: Exam Access Granted

```
Output    Serial Monitor ×
Message (Enter to send message to 'Arduino UNO' on 'COM16')
Card Scanned...
Name: Aaryaman Singh
Roll Number: 24MCA20449
Exam Access Granted
----------------------
Exam Hall Access System Ready...
Card Scanned...
Name: Dimpal Tyagi
Roll Number: 24MCA20021
Exam Access Granted
----------------------
Card Scanned...
Name: Ankush Koundal
Roll Number: 24MCA20251
Exam Access Granted
----------------------
```

UID not matched: Not Eligible for exam

```
----------------------
Card Scanned...
Not Eligible for Exam
----------------------
Card Scanned...
Not Eligible for Exam
----------------------
```

4. Learning outcomes (What I have learnt):

1. Understand how RFID technology works for identification and access control.

2. Learn how to read RFID card UID using MFRC522 module and Arduino.

3. Control output devices (like buzzer) based on access granted or denied**.**

4. Implement decision making by comparing scanned UID with stored data.